Boundary Stability for Adaptive Stopping in Diffusion-Based Event Boundary Detection

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Abstract

Diffusion-based generic event boundary detection (GEBD) performs multiple denoising steps, yet a small mean change in the scores alone does not imply that the detected boundaries have stabilized. We analyze the mismatch between score changes and boundary changes in a pretrained DiffGEBD and evaluate a training-free method that selects a per-video stopping step from the number and locations of boundaries at adjacent steps. Among the 18,797 available validation videos of Kinetics-GEBD, 55.97% exhibit boundary changes despite small mean score changes, and among the 18,447 videos subject to the official evaluation, 14.29% exhibit changes in ground-truth matching or in TP/FP/FN. On 12,900 confirmation videos whose IDs are disjoint from the policy-selection split, the proposed method improves F1@0.05 by 0.635 percentage points (pp) over fixed 4-step DDIM while reducing batch-1 model inference time on an RTX 4060 Ti by 5.23%. In 1,000 controls that randomly reassign the same distribution of stopping steps across videos, the F1 of the proposed method is 0.113 pp above the reassignment mean, indicating a conditional gain from per-video allocation of computation. These results, however, are limited to an already-observed validation set and a single sampling trajectory; they imply neither superiority over every fixed stopping point on the same schedule nor a guarantee that the accuracy of the full computation is preserved.

1 Introduction

Generic event boundary detection (GEBD) localizes semantically meaningful transition points in a video without predefining action categories. Kinetics-GEBD provides the judgments of multiple annotators on changes in scene, action, object, and the like (Shou et al., 2021). To account for the subjectivity of boundaries, DiffGEBD generates boundary predictions by iteratively denoising conditioned on video features (Hwang et al., 2025). We keep the weights of this model frozen and focus on when to halt the iterative computation.

Stability in score space is distinct from stability of the final detections. Even when the mean change over the 100 positions is small, a boundary segment can appear or vanish, or its center can shift, if the scores at a few positions cross the detection threshold. Conversely, even if the predictions keep changing, the final step is not always more accurate. There is therefore room to determine the amount of computation from the current detections rather than applying the same number of steps to every video. This possibility must be distinguished from any claim that guarantees boundary invariance after stopping or an error bound.

We examine a rule that returns the current prediction when two adjacent steps yield the same number of boundaries and the maximum displacement between corresponding locations falls within a tolerance. The stopping decision uses only the current and immediately preceding predictions and has no access to the ground truth

or to future steps. In our validation, we distinguish between truncating the same 32-step trajectory midway and fixed few-step DDIM whose schedule is respaced from the outset. We further assess the correspondence between videos and computation itself through random reassignment that preserves the distribution of stopping steps.

This work presents three pieces of empirical evidence. First, we quantify the phenomenon in which boundaries and the official ground-truth matching change even when the mean score is stable. Second, with an implementation that actually skips the subsequent computation, we confirm improvements in both accuracy and model latency over fixed 4-step DDIM. Third, we show that per-video assignment of stopping steps yields a small accuracy gain even under an identical distribution of computation steps. Our contribution is a controlled empirical analysis of a single public checkpoint; we do not claim universal optimality of the stopping rule or generalization to new independent data.

2 Related Work

Representation and efficiency in GEBD. DDM-Net represents temporal changes using dense difference maps of multi-level features and progressive attention (Tang et al., 2022). EfficientGEBD revisits the accuracy and computational cost of boundary detection architectures (Zheng et al., 2024b). DyBDet employs multiple exits and dynamic processing per video segment (Zheng et al., 2024a). Unlike these works, we leave the trained diffusion model untouched and terminate the denoising iterations for the entire video. We do not conduct experiments that prune the features at specific positions or restore computation for neighboring boundary candidates.

Diffusion-based boundary detection and computation reduction. DiffGEBD casts the diversity of boundary annotations as a generation problem and controls the diversity and condition fidelity of its predictions with classifier-free guidance (CFG) (Hwang et al., 2025). DDIM provides the basis for reducing the number of steps by altering the inference schedule (Song et al., 2021). AdaptiveDiffusion studies the reuse of noise predictions based on the third-order latent difference across timesteps (Ye et al., 2024), and SADA adjusts step-wise and token-wise sparsification according to trajectory stability (Jiang et al., 2025). We therefore do not claim stability-driven computation reduction itself as a novel idea. Our focus lies in using the post-processed boundary set of GEBD as the stopping signal and in disentangling the effect of the schedule from that of per-video computation allocation. A direct performance comparison in which these general-purpose accelerators are ported to our model is not included.

3 Boundary Stability Based Stopping

3.1 Score Change versus Boundary Change

We denote by $q(v,t)\in[0,1]^L$ the clean prediction obtained at denoising step t for video v, where $L=100$. Note that q is not the noisy state but the score after CFG and clipping. The mean absolute change between adjacent steps is given by

$$\Delta_{v,t} = \frac{1}{L}\sum_{i=1}^{L} |q_{v,t,i} - q_{v,t-1,i}|.$$

The official post-processing Φ converts contiguous segments above the detection threshold of 0.5 into boundary center coordinates. We define $B(v,t)=\Phi(q(v,t))$ as the boundary set sorted in ascending order. Diagnostic H1a refers to the case in which B changes at a transition at which $\Delta \le 0.01$

has held for three consecutive transitions. For diagnostic purposes, boundaries are matched by a maximum-cardinality, minimum-distance one-to-one assignment within a radius of 5% of the video length; unmatched boundaries are classified as births or deaths, and matched boundaries whose coordinates change by more than 0.5 frames as shifts. H1b refers to the case in which, under the same condition, the official TP/FP/FN or the set of matched GT boundaries of the same annotator changes. This is a post-hoc diagnostic that uses the ground truth and is kept separate from the inputs to the stopping rule.

3.2 Per-Video Stopping Rule

The proposed boundary-stability-based stopping directly compares the number and locations of the boundaries in the two sets. If the two sets have the same non-zero cardinality, their elements are matched one-to-one in ascending order and the maximum displacement d(v,t) is computed. We define d=0 if both sets are empty and d=∞ if their cardinalities differ.

$$d_{v,t} = \max_{1 \le j \le |B_{v,t}|} |b_{v,t,j} - b_{v,t-1,j}|.$$

Inference stops at the first step at which d≤ε has held for h consecutive transitions. The selected configuration is ε=3 annotation frames and h=1, which can be written as follows.

$$T(v) = \min\left(\{t : d_{v,t} \le 3\} \cup \{32\}\right).$$

Here t∈{2,...,32}, so the earliest possible stop is at step 2. If the condition is never satisfied, computation proceeds through step 32. The returned value is the clean prediction at the stopping step, and the next DDIM update and decoder call are not executed. Video features are computed once, and only the current and preceding boundary sets are retained. Because the unit is annotation frames, the duration corresponding to 3 frames depends on the video fps. This rule is an empirical criterion for allocating iterative computation; it provides no guarantee that the boundaries remain unchanged after stopping or that the current prediction is correct.

3.3 Factors Controlled in the Comparisons

Fixed early stopping executes only the first k steps of the same 32-step schedule. In contrast, respaced k-step DDIM uses a schedule constructed for k total steps from the outset. For example, the timesteps of fixed 4-step early stopping are [999,967,936,905], whereas those of respaced 4-step DDIM are [999,749,499,249]. Full 32-step inference runs the original trajectory to completion. Without this distinction, the effect of changing the schedule could be mistaken for the effect of adaptivity. In the adaptivity control, we reassign the per-video list of stopping steps produced by boundary-stability-based stopping via a random permutation and evaluate each video's prediction at its assigned step. Since the total number of steps and the number of videos per step are identical, the effect of the video–stopping-step correspondence can be compared conditionally.

4 Experiments

4.1 Data and Implementation

Of the 18,815 official Kinetics-GEBD validation annotations, we use the 18,797 videos in our possession. The 18 videos with missing source files were excluded, and no execution failures occurred. Accuracy is aggregated over the 18,447 videos satisfying the official consistency≥0.3 criterion. After an exploratory diagnosis on the full set, we formed a selection split of 5,639 videos and a confirmation split of 13,158 videos using a fixed 30/70 partition with seed 20261012; the officially evaluated videos number 5,547 and 12,900,

respectively. Although the confirmation split has no ID overlap with the selection split, it was already observed during the diagnosis and is therefore not a new independent test set.

The input consists of 100 uniformly sampled RGB frames at 224×224, processed by ResNet-50 layer4 and a 6-layer Transformer. We retain 32-step DDIM, $\eta=0$, the CFG formula $c+7(c-u)$ as implemented in the code, a detection threshold of 0.5, FP32, and disabled AMP/TF32. A single sampling trajectory per video is used with a seed derived from the video ID, and no retraining is performed.

Because we resize the original RGB frames directly, pixel-level identity with the authors' JPEG cache could not be verified. For the 43 videos whose source and annotation frame counts differ, correspondence was established via normalized video positions. The F1@0.05 over the full evaluation set was 78.048%. Hwang et al. (2025, Table 4) report 78.4% at a CFG weight of $w=4.0$. Since our experiments use $w=7$ together with a different preprocessing pipeline and set of available videos, this figure is given as a reference value rather than as a reproduction under identical conditions. All subsequent comparisons between methods use the same checkpoint, preprocessing, videos, and initial noise.

4.2 Policy Selection and Evaluation Protocol

The candidates for boundary-stability-based stopping are the 12 combinations of $\varepsilon\in\{0.5,1,2,3\}$ and $h\in\{1,3,5\}$. On the selection split, for each tolerated F1 drop of 0/0.1/0.3 pp relative to full 32-step inference, we selected the candidate with the smallest mean number of steps, breaking ties by F1 and then by lexicographic order. The configuration $\varepsilon=3$, $h=1$ was selected under all three budgets. The policy was not modified after observing the confirmation results. We also compared mean-score-change-based stopping and mean-and-max-score-change-based stopping over $\tau\in\{0.001,0.005,0.01,0.02\}$ and $h\in\{1,3,5\}$. The former stops once h consecutive transitions have a mean change of at most τ, while the latter additionally requires the maximum change to be at most the same τ. In the latter, the max-change condition subsumes the mean-change condition, so it is a restricted control in which the maximum change effectively determines stopping.

The primary accuracy metric is the conventional F1@0.05. Following the official code, the annotator yielding the highest per-video F1 is selected and TP/FP/FN are summed over videos. Ties are resolved in favor of the first annotator, annotator 0 is used for empty predictions, and nearest-unused matching in GT order is retained. This metric is distinct from the mean of per-video F1 and from the diversity-aware symmetric F1 of DiffGEBD. Paired percentile 95% confidence intervals (CIs) were computed from 5,000 video-level bootstrap resamples drawn with replacement. These intervals are conditional on the fixed policy; they do not incorporate variability due to policy selection or seeds, and no correction was applied for the auxiliary multiple comparisons.

Wall-clock time was measured on an RTX 4060 Ti 16GB at batch size 1, with three runs for each of 200 confirmation videos chosen independently of the predictions. The measurement spans from the point at which the input tensor is ready on the GPU to the return of the final boundaries on the CPU, and includes the backbone, decoder, stopping decision, post-processing, and synchronization. We applied warm-up and alternated the order of the methods, and performed a paired bootstrap on the per-video mean of the three runs. Decoding, input transfer,

and file saving are excluded from the primary timing. For the primary comparison between boundary-stability-based stopping and respaced 4-step DDIM, we pre-specified the criteria that the lower CI bound of the F1 difference be >-0.1 pp and that the upper CI bound of the latency ratio be <1. The 0.1 pp margin is an operational criterion of this study.

4.3 Changes Missed by Mean-Score Stability

H1a was observed in 10,521/18,797 videos (55.972%) and H1b in 2,636/18,447 (14.290%). In a post-hoc analysis with the previous annotator held fixed, both recoveries (528 cases) and losses (1,885 cases) of ground-truth matches were present. Every H1a transition involved a threshold crossing, and the median of the maximum score change was 0.16358. When MAX≤0.01 was added to the MAE condition, no H1b cases were observed, but this condition applied to only 1,478 transitions in 454 videos. Thus, while the dilution of local changes by averaging is observed, this fact alone does not warrant the conclusion that a boundary-based criterion is necessary or superior to all score-based criteria.

4.4 Accuracy and Wall-Clock Time of Actual Stopping

Table 1. Accuracy and wall-clock model time on the confirmation split. Accuracy: n=12,900; time: n=200 videos×3 runs. All results were measured in the same session; F1 is in % and time in ms. Abbreviations: D=boundary-stability-based stopping, P_k=fixed k-step early stopping on the 32-step trajectory, S_k=respaced k-step DDIM, F_{32}=full 32-step inference.

Method	Avg. steps	F1 ↑	Time ↓
D	2.416	77.956	250.39
P_3	3	77.990	254.98
P_4	4	78.185	263.73
S_4	4	77.321	264.20
S_8	8	77.779	299.53
F_{32}	32	78.024	505.14

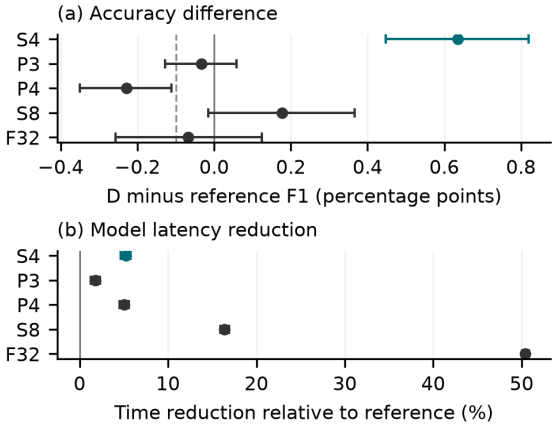


Figure 1. F1 difference (top) and time reduction (bottom) of boundary-stability-based stopping (D) relative to each reference. P_k=fixed k-step stopping on the 32-step trajectory, S_k=respaced k-step DDIM, F_{32}=full inference. Same measurement session as Table 1; bars denote paired 95% CIs and the dashed line marks the tolerated loss of −0.1 pp.

As shown in Table 1, boundary-stability-based stopping terminates after 2.416 steps on average, achieving an F1 that is 0.6353 pp higher than that of respaced 4-step DDIM (CI [0.4473,0.8193]) and a model latency that is 5.23% lower (CI [4.67,5.76]). The output of the actual early-stopped run

matched the stored scores at the corresponding step with a maximum error of 0, and no additional calls were made after stopping. Relative to respaced 4-step DDIM, the number of iterative calls decreased by about 39.6%, but because the remaining cost, including the backbone, is about 222 ms, the reduction in total model time was small. This is not a figure for end-to-end pipeline throughput or for service-level speedup including decoding.

Relative to respaced 8-step DDIM, the F1 difference of boundary-stability-based stopping is +0.1775 pp (CI [−0.0155,0.3659]); an improvement as such cannot be established, but both the pre-specified tolerated-loss criterion and the time-reduction criterion (16.41%) are met. Relative to full 32-step inference, a 50.43% time reduction was observed, but the CI [−0.2577,0.1239] of the F1 difference of −0.0679 pp extends below −0.1 pp, so the accuracy-preservation criterion is not met. Compared with fixed 4-step early stopping, boundary-stability-based stopping is faster but 0.2291 pp lower in F1, so superiority over every fixed stopping point on the same schedule cannot be claimed.

In an offline comparison with score-based criteria, boundary-stability-based stopping attained an F1 0.1437 pp higher than the fast configuration of mean-score-change-based stopping (τ=0.02, h=1) (CI [0.0583,0.2301]) with a smaller mean number of steps (2.416 vs. 2.579). The configuration of mean-and-max-score-change-based stopping chosen on the selection split yielded 31.083 steps on average and an F1 of 78.030%. These results constitute an accuracy–step comparison over a finite candidate range, and the actual latency of the two score-based stopping methods was not measured. Even after boundary-stability-based stopping, boundary changes occurred in 99.233% of the evaluated videos and H1b in 86.271%. Since these changes include both improvements and degradations, we do not interpret local stability as invariance of the future trajectory.

4.5 Adaptivity under an Identical Step Distribution

On the 12,900 confirmation videos, the 31,166 total steps of boundary-stability-based stopping (mean 2.415969) were reassigned 1,000 times with seed 20261015. Every permutation exactly preserves the step histogram and the total, and the official annotator selection and evaluation were re-applied at each video's assigned step. The identity assignment, which retains the original video–step correspondence, coincided with boundary-stability-based stopping. This control was computed from stored intermediate outputs, and GPU time is not assumed to be equal.

Table 2. Reassignment control under the same step distribution. The range is given by quantiles of the reassignment distribution and is not a confidence interval of performance. D denotes boundary-stability-based stopping.

Item	F1
D	77.95615%
Reassignment mean	77.84351%
Reassignment 2.5–97.5% range	77.78852–77.89442%
D − reassignment mean	+0.11264%p
Reassignments ≥ D	0 / 1,000

Boundary-stability-based stopping attained a higher F1 than every reassignment, with a difference of +0.11264 pp from the reassignment mean (Table 2). The one-sided Monte Carlo proportion, (1+number of reassignments with F1 at least that of the proposed method)/1001, is 0.000999. This constitutes evidence, conditional on the fixed set and step distribution, under the null

hypothesis that the correspondence between videos and stopping steps is exchangeable. The 1,000 permutations are not independent samples of videos, and the reassignment quantile interval is not a CI of generalization performance. Although the effect is small, it supports a gain from per-video assignment that is not explained by the total number of iterations alone.

Table 3. Separate concurrent measurement against fixed 2- and 3-step early stopping. P_2 and P_3 execute only the first 2 and 3 steps of the 32-step trajectory, respectively. Each difference is boundary-stability-based stopping minus the reference, with paired 95% CIs in brackets.

Reference	ΔF1 (pp)	Time reduction (%)
P_2	+0.183 [+0.116,+0.249]	−2.13 [−2.70,−1.58]
P_3	−0.034 [−0.128,+0.058]	+1.58 [+1.01,+2.11]

In a separate concurrent measurement against simple alternatives, boundary-stability-based stopping was 0.1826 pp more accurate than fixed 2-step early stopping but 2.13% slower. It was 1.58% faster than fixed 3-step early stopping, but the CI of the F1 difference straddled −0.1 pp (Table 3). We therefore interpret its relation to fixed 2- and 3-step early stopping as an accuracy–time trade-off. The absolute times in Tables 1 and 3 were obtained in different measurement sessions, so we do not compute differences across them. The reassignment gain of 0.113 pp and the 0.635 pp gain over respaced 4-step DDIM are relative to different references and are thus neither summed nor decomposed into per-cause contributions.

5 Discussion and Limitations

The current evidence is limited to a single checkpoint, one sampling trajectory per video, an already-observed validation set, and batch size 1 on a single GPU. The confirmation split is not a new independent test set, and the bootstrap does not remove the uncertainty of policy selection. A fix to the computation-graph connectivity, direct RGB preprocessing, and some of the coordinate correspondences remain as differences from the original authors' runs. The fps-dependent displacement tolerance and the rule that treats two empty boundary sets as stable are further limitations to generalization.

This work does not establish universal optimality of boundary-based stopping, preservation of future boundaries, superiority over every fixed stopping point, or acceleration across multiple devices. Nor does the control matched on the step-count distribution equalize the decision overhead. Future work requires re-validation with the policy frozen on new evaluation data and multiple noise seeds, as well as direct wall-clock comparisons with other simple adaptive criteria. The present results serve as empirical grounds for such validation and are not extended to broad stability guarantees.

6 Conclusion

We analyzed the mismatch between mean-score stability and detected-boundary stability in diffusion-based GEBD and evaluated a training-free adaptive stopping method that uses the number and locations of boundaries. Both F1 and model inference time improved over fixed 4-step DDIM, and a reassignment control under the same step distribution confirmed a small accuracy gain from per-video selection of the stopping step. However, a trade-off remained with respect to fixed stopping on the same schedule, and the accuracy-preservation criterion relative to full computation was not established. Our results support the potential of allocating computation

according to boundary structure, while calling for validation on independent data and additional sampling trajectories.

Impact Statement

This work concerns an inference method that reduces the iterative computation of video event boundary detection. Reduced computation may lower the resource usage of video analysis, but energy savings were not measured directly. Since missed boundaries and differences in subjective event segmentation can affect video summarization or retrieval results, the accuracy–time relationship observed in our experiments must not be used as a guarantee for safety-critical decision-making. When using public video data, the terms of use of the original data and privacy protection must be observed.

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